

Georges Le Bellier

Ph.D. Student - Generative Deep Learning

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Professional Summary

I am a third-year Ph.D. student at CNAM in Paris, and I research generative models (diffusion and flow-matching models) and data-to-data translation methods to develop robust pipelines and enable pixel-space domain adaptation for satellite imagery.

Professional Experience

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| 01/2023
Present | - | Conservatoire National des Arts et Métiers (Ph.D)
<i>I research generative models and domain adaptation for Earth Observation; focusing on flow-based methods (diffusion and flow-matching models) and data-to-data translation (diffusion and Schrödinger bridges), targeting remote sensing applications.</i> |
| 04/2022
10/2022 | - | SONY Computer Science Lab (Research Intern)
<i>I explored advanced diffusion generative models and designed conditional generation pipelines to provide enhanced controllability for musicians in AI-driven music creation.</i> |
| 03/2021
08/2021 | - | IRCAM / Pixtunes GmbH (Research Intern)
<i>I designed and implemented multiple generative deep learning algorithms for expressive MIDI-to-audio translation. I focused on autoregressive and diffusion models for pitch and loudness timeseries generation.</i> |
| 02/2019
03/2019 | - | INRIA (Research Intern)
<i>I carried out an internship in the LINKS computer science research team from INRIA laboratories and wrote a bibliographic research review on this topic.</i> |

Main publications and projects

1. **FlowEO: Generative Unsupervised Domain Adaptation for Earth Observation**, Georges Le Bellier, and Nicolas Audebert. (Preprint 2025)
2. **Riemannian Flow Matching for SAR Interferometry**, Georges Le Bellier, Dana El Hajjar, Arnaud Breloy, and Nicolas Audebert. (GRETSI 2025)
3. **PANGAEA: A Global and Inclusive Benchmark for Geospatial Foundation Models**, V. Marsocci, Y. Jia, G. Le Bellier, D. Kerekes, L. Zeng, S. Hafner, S. Gerard, E. Brune, R. Yadav, A. Shibli, H. Fang, Y. Ban, M. Vergauwen, N. Audebert, A. Nascetti (Preprint 2015)
4. **Detecting Out-Of-Distribution Earth Observation Images with Diffusion Models**, Georges Le Bellier, and Nicolas Audebert. (EarthVision CVPR 2024)
5. **Educational notebooks on flow matching**, Georges Le Bellier. I developed educational implementations and detailed explanations of fundamental ideas from flow matching and discrete flow matching research articles.

Education

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| 2018 - 2022 | | École Centrale de Lille , Lille
<i>M.Sc. SDIA - machine learning, deep learning, signal processing, optimization.</i> |
| 2020 - 2021 | | Sorbonne University/IRCAM/ParisTech , Paris
<i>M.Sc. ATIAM - signal processing, deep learning, generative modeling, acoustics.</i> |
| 2016 - 2018 | | Lycée Victor Grignard , Cherbourg
<i>CPGE MP - 2-year advanced undergraduate studies in mathematics, physics, computer science.</i> |